THE KUBERT–TATE NORMAL FORM AND GOOD REDUCTION EVERYWHERE, WITH SOLUTIONS

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This worksheet is about two sections of [Sil09]: torsion points (§VIII.7), and minimal equations and good reduction (§VIII.8). Mazur's theorem lists the fifteen groups that occur as the torsion subgroup of an elliptic curve over $\mathbb{Q}$. The Kubert–Tate normal form puts an elliptic curve with a k-point P of order $m \geqslant 4$ in a family with two parameters, in which m is cut out by a polynomial condition on the parameters. These conditions define curves, the modular curves $Y_1(m)$, whose k-points parametrize such pairs (E, P). Problem 1.4 derives the normal form, and finds the conditions for $m = 4, 5, 6, 7, 9, 11, 12$. For $m = 11$ the curve has genus one. Problem 2.5 shows that every elliptic curve over $\mathbb{Q}$ has bad reduction at some prime. Each problem is preceded by the definitions and results it uses.

Standing notation.

- A field k, and an elliptic curve E over k given by a Weierstrass equation
$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$
with coefficients in k, discriminant Δ and point at infinity O [Sil09, §III.1].
- A change of variables $x = u^2x' + r$, $y = u^3y' + su^2x' + t$, with $u \in k^\times$ and $r, s, t \in k$. It gives a Weierstrass equation for E in x', y', with discriminant $u^{-12}\Delta$. Any two Weierstrass equations for E are related by a change of variables [Sil09, Proposition III.3.1(b)].
- Coordinates $x(Q)$ and $y(Q)$ of a $\overline{k}$-point $Q \neq O$ of E.

How to use this. Allow about 50 minutes for Problem 1.4 and 20 for Problem 2.5. Magma is needed only in Problem 1.4(9). Everything else is by hand.

1. The Kubert–Tate normal form and Mazur's list

Problem 1.4 is [Sil09, Exercise 8.13]. Its item (1) is Exercise 8.13(a), the orders 6, 7, 9 and 12 are Exercise 8.13(b), and the order 11 is Exercise 8.13(c). The normal form, and the conditions for all the orders allowed by Theorem 1.1, are in [Kub76]. The elliptic curves with a k-point of order $m \geqslant 4$ are parametrized by the k-points of a modular curve [Sil09, Remark VIII.7.8].

Theorem 1.1 (Mazur [Sil09, Theorem VIII.7.5])**.** *Let E be an elliptic curve over $\mathbb{Q}$. Then $E(\mathbb{Q})_{\mathrm{tors}}$ is isomorphic to one of the fifteen groups*
$$\mathbb{Z}/N\mathbb{Z} \text{ with } 1 \leqslant N \leqslant 10 \text{ or } N = 12, \qquad \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2N\mathbb{Z} \text{ with } 1 \leqslant N \leqslant 4.$$
Each of them occurs as $E(\mathbb{Q})_{\mathrm{tors}}$ for some elliptic curve E over $\mathbb{Q}$.

Proposition 1.2 (The group law [Sil09, Proposition III.2.2(a) and Algorithm III.2.3(a)])**.** *If a line in $\mathbb{P}^2$ meets E at the $\overline{k}$-points P_1, P_2, P_3, counted with multiplicity, then $P_1 + P_2 + P_3 = O$. The negative of a $\overline{k}$-point (x_0, y_0) is $(x_0, -y_0 - a_1x_0 - a_3)$.*

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Definition 1.3 (Kubert–Tate normal form)**.** A Weierstrass equation is in Kubert–Tate normal form if it is

$$y^2 + (1-c)xy - by = x^3 - bx^2 \tag{1.1}$$

for some $b, c \in k$.

Problem 1.4 (The Kubert–Tate normal form and Mazur's list)**.**

(1) Let $P \in E(k)$ be a k-point of order $m \geqslant 4$, possibly infinite. In three steps, show that a change of variables gives a Weierstrass equation for E in Kubert–Tate normal form, in which $P = (0,0)$. After each change of variables, write again x, y for the coordinates and $a_1, \dots, a_6$ for the coefficients.

(a) Show that a change of variables with $u = 1$ and $s = 0$ moves P to $(0,0)$, and that then $a_6 = 0$.

SOLUTION. **Take $r = x(P)$ and $t = y(P)$.** Since $P \neq O$, it is an affine k-point $(x(P), y(P))$. The change of variables $x = x' + x(P)$, $y = y' + y(P)$ has $u = 1$ and $s = 0$, and gives P the coordinates $(0,0)$. As $(0,0)$ satisfies the new equation, its constant term a_6 is 0.

(b) Suppose that $P = (0,0)$, so that $a_6 = 0$. Show that $a_3 \neq 0$, and that the change of variables with $u = 1$, $r = t = 0$ and $s = a_4/a_3$ makes $a_4 = 0$. Show that then the tangent line to E at P is $y = 0$.

SOLUTION. **If $a_3 = 0$, the tangent line at P is $x = 0$, and then $2P = O$.** Put

$$F := y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x.$$

At $(0,0)$, $\partial F/\partial x$ takes the value $-a_4$ and $\partial F/\partial y$ the value a_3, and they are not both 0, since E is nonsingular. So the tangent line at P is $a_3y = a_4x$. If $a_3 = 0$, it is the line $x = 0$. In projective coordinates it is $X = 0$, which passes through $O = [0:1:0]$. So it meets E at P twice and at O, and Proposition 1.2 gives $2P = O$, a contradiction. So $a_3 \neq 0$. Substituting $y = y' + sx$ gives

$$y'^2 + (a_1 + 2s)xy' + a_3y' = x^3 + (a_2 - sa_1 - s^2)x^2 + (a_4 - sa_3)x,$$

whose coefficient of x is 0 for $s = a_4/a_3$. The new coordinates of P are still $(0,0)$, and the tangent line there is $a_3y' = 0$, that is, $y' = 0$.

(c) Suppose that $P = (0,0)$ and $a_4 = a_6 = 0$, with $a_3 \neq 0$. Show that $a_2 \neq 0$, and that the change of variables with $u = a_3/a_2$ and $r = s = t = 0$ gives a Weierstrass equation in Kubert–Tate normal form. Find b and c.

SOLUTION. **Here $b = -a_2^3/a_3^2$ and $c = 1 - a_1a_2/a_3$.** As in (1)(b), the tangent line at P is $a_3y = a_4x$, that is, $y = 0$. On it the equation becomes $x^3 + a_2x^2 = x^2(x + a_2) = 0$. If $a_2 = 0$, the line meets E at P three times, and Proposition 1.2 gives $3P = O$, a contradiction. So $a_2 \neq 0$, and $u := a_3/a_2$ lies in $k^\times$. Substituting $x = u^2x'$ and $y = u^3y'$, and dividing by u^6, gives

$$y'^2 + \frac{a_1}{u}x'y' + \frac{a_3}{u^3}y' = x'^3 + \frac{a_2}{u^2}x'^2.$$

Here $a_3/u^3 = a_2^3/a_3^2 = a_2/u^2$ and $a_1/u = a_1a_2/a_3$. So the equation is in Kubert–Tate normal form, with $b = -a_2^3/a_3^2$ and $c = 1 - a_1a_2/a_3$, and P is still $(0,0)$. The composite of the three changes of variables is again a change of variables, which proves (1).

(2) Conversely, suppose that E is given by an equation in Kubert–Tate normal form. Let $P := (0,0) \in E(k)$, and let m be its order, possibly infinite. Show that $b \neq 0$, that $2P = (b, bc)$ and $3P = (c, b-c)$, and that $m \geqslant 4$.

Solution. **The tangent line at P is $y = 0$, and the line through P and $2P$ is $y = cx$.** If $b = 0$, every term of the equation has degree at least 2 in x and y, so $(0,0)$ is a singular point, which is impossible. So $b \neq 0$. As in (1)(b), the tangent line at P is $a_3 y = a_4 x$, here $-by = 0$, that is, $y = 0$. On it the equation becomes $x^3 - bx^2 = x^2(x-b) = 0$. So the line meets E at P twice and at $(b,0)$, and Proposition 1.2, with $a_1 = 1-c$ and $a_3 = -b$, gives

$$2P = -(b,0) = \big(b, -(1-c)b + b\big) = (b, bc).$$

As $b \neq 0$, the line through P and $2P$ is $y = cx$. On it the equation becomes

$$c^2x^2 + (1-c)cx^2 - bcx = x^3 - bx^2, \quad \text{that is,} \quad x(x-b)(x-c) = 0.$$

So the line meets E at P, $2P$ and (c, c^2), counted with multiplicity (which matters when $c = 0$ or $c = b$), and

$$3P = -(c, c^2) = \big(c, -c^2 - (1-c)c + b\big) = (c, b-c).$$

The point at infinity O is none of P, $2P$ and $3P$, which are affine. So $m \geqslant 4$.

(3) With E and P as in (2), suppose that a change of variables gives another equation for E in Kubert–Tate normal form, with parameters b' and c', in which P is again $(0,0)$. Show that $b' = b$ and $c' = c$. So the pair (E, P) determines b and c.

Solution. **The change of variables is the identity.** As P is $(0,0)$ in both coordinates, $r = t = 0$. Substituting $x = u^2x'$ and $y = u^3y' + su^2x'$ in (1.1), and dividing by u^6, gives as in (1)(b)

$$a_1' = \frac{a_1 + 2s}{u}, \quad a_2' = \frac{a_2 - sa_1 - s^2}{u^2}, \quad a_3' = \frac{a_3}{u^3}, \quad a_4' = \frac{a_4 - sa_3}{u^4}.$$

Both equations have $a_4 = a_4' = 0$, and $a_3 = -b \neq 0$ by (2). So $s = 0$. Both have $a_2 = a_3$ and $a_2' = a_3'$, so $a_2/u^2 = a_2/u^3$ with $a_2 = -b \neq 0$, and $u = 1$. Then $b' = -a_3' = b$, and $1 - c' = a_1' = a_1 = 1 - c$.

(4) With E, P and m as in (2), show that $m = 4$ if and only if $c = 0$, and $m = 5$ if and only if $b = c$.

Solution. **Here $m = 4$ means $3P = -P$, and $m = 5$ means $3P = -2P$.** By (2), $m \geqslant 4$. So $m = 4$ if and only if $4P = O$, that is, $3P = -P$. By Proposition 1.2, $-P = (0,b)$, and $3P = (c, b-c)$ equals $(0,b)$ if and only if $c = 0$. Likewise, $m = 5$ if and only if $5P = O$, since 5 is prime, that is, $3P = -2P$. Here $-2P = (b,0)$, the third k-point on the tangent line at P, and $(c, b-c) = (b,0)$ if and only if $c = b$.

Remark. The formulas of [Sil09, §III.1] give the discriminant of (1.1):

$$\Delta = b^3\big(16b^2 - (8c^2 + 20c - 1)b + c(c-1)^3\big).$$

For $c = 0$ it is $b^4(16b+1)$, and for $b = c$ it is $b^5(b^2 - 11b - 1)$. So, up to a change of variables, the elliptic curves with a k-point of order 4 are the curves $y^2 + xy - by = x^3 - bx^2$ with $b(16b+1) \neq 0$, and those with a k-point of order 5 are the curves $y^2 + (1-b)xy - by = x^3 - bx^2$ with $b(b^2 - 11b - 1) \neq 0$. For $b = 1$ the second family gives $y^2 - y = x^3 - x^2$, with $\Delta = -11$, from [Sil09, Exercise 8.12(e)].

(5) With E, P and m as in (2), find the condition on b and c for $m = 6$.

Solution. **Here $b = c + c^2$.** By (2), $m \geqslant 4$, so $m = 6$ if and only if $6P = O$, that is, $3P = -3P$. By Proposition 1.2, $-3P = (c, c^2)$, the third k-point on the line through P and $2P$. And $(c, b-c) = (c, c^2)$ if and only if $b = c + c^2$.

Remark. For $b = c + c^2$ the discriminant is $c^6(c+1)^3(9c+1)$. So, up to a change of variables, the elliptic curves with a k-point of order 6 are the curves

$$y^2 + (1-c)xy - (c+c^2)y = x^3 - (c+c^2)x^2, \qquad c(c+1)(9c+1) \neq 0.$$

For $c = 1$ this is $y^2 - 2y = x^3 - 2x^2$, with $\Delta = 80$.

(6) With E, P and m as in (2), suppose that $m \geqslant 5$, so that $c \neq 0$, and put $d := b/c$. Show that

$$4P = \big(d^2 - d,\ d^2(c - d + 1)\big).$$

Solution. **The line through P and $3P$ meets E again at $-4P = (d^2 - d, d(d-1)^2)$.** Here $b = cd$ and $3P = (c, c(d-1))$, so the line through P and $3P$ is $y = (d-1)x$. On it the equation becomes

$$x^3 - \big((d-1)^2 + (1-c)(d-1) + cd\big)x^2 + cd(d-1)x = 0.$$

Its roots are 0, c and a third root x_3, counted with multiplicity, and $cx_3 = cd(d-1)$. As $c \neq 0$, $x_3 = d(d-1)$. By Proposition 1.2 the third k-point on the line is $-(P + 3P) = -4P = (d(d-1), d(d-1)^2)$, and

$$4P = \big(d(d-1),\ -d(d-1)^2 - (1-c)d(d-1) + cd\big) = \big(d^2 - d,\ d^2(c - d + 1)\big).$$

(7) With $m \geqslant 5$ and d as in (6), show that $m = 7$ if and only if P, $2P$ and $4P$ are collinear, and that this holds if and only if $c = d^2 - d$. Deduce a one-parameter family of elliptic curves with a k-point of order 7.

Solution. **Here $c = d^2 - d$ and $b = d^3 - d^2$, with $d \in k$.** The k-points P, $2P$ and $4P$ are distinct, since P, $2P$ and $3P$ are not O. The line through P and $2P$ meets E again at $-3P$. If $4P$ lies on it, then $4P = -3P$, so $7P = O$ and $m = 7$, as 7 is prime. Conversely, if $m = 7$, then $-3P = 4P$ lies on the line. By (2) this line is $y = cx$, and by (6) the k-point $4P$ lies on it if and only if $d^2(c - d + 1) = cd(d-1)$. As $d \neq 0$, this is $dc - d^2 + d = cd - c$, that is, $c = d^2 - d$. Then $b = cd = d^3 - d^2$.

Remark. This is the family of [Sil09, Remark VIII.7.8]:

$$y^2 + (1 + d - d^2)xy + (d^2 - d^3)y = x^3 + (d^2 - d^3)x^2, \qquad \Delta = d^7(d-1)^7(d^3 - 8d^2 + 5d + 1).$$

(8) With d as in (6), suppose that $m \geqslant 6$, so that $d \neq 1$, and put $f := c/(d-1)$. Show that $m = 9$ if and only if $2P$, $3P$ and $4P$ are collinear, and that this holds if and only if

$$c^2 - (d-1)c - (d-1)^3 = 0, \quad \text{that is,} \quad d = f^2 - f + 1.$$

Deduce a one-parameter family of elliptic curves with a k-point of order 9.

Solution. **Here $d = f^2 - f + 1$, $c = f^2(f-1)$ and $b = f^2(f-1)(f^2 - f + 1)$, with $f \in k$.** The k-points $2P$, $3P$ and $4P$ are distinct, since P and $2P$ are not O. The line through $2P$ and $3P$ meets E again at $-5P$. If $4P$ lies on it, then $4P = -5P$, so $9P = O$ and $m = 9$, as $m \geqslant 4$. Conversely, if $m = 9$, then $-5P = 4P$ lies on the line. Here $2P = (cd, c^2 d)$ and $3P = (c, c(d-1))$, so the line has slope $\lambda = (cd - d + 1)/(d-1)$, and $4P$ lies on it if and only if $(d-1)\big(y(4P) - y(3P)\big) = (d-1)\lambda\big(x(4P) - x(3P)\big)$, that is,

$$(d-1)\big(d^2(c - d + 1) - c(d-1)\big) = (cd - d + 1)(d^2 - d - c).$$

Expanding, the difference of the two sides is $d\big(c^2-(d-1)c-(d-1)^3\big)$, and $d\neq 0$. Substituting $c=f(d-1)$ gives $(d-1)^2\big(f^2-f-(d-1)\big)=0$, that is, $d=f^2-f+1$. Then $c=f(d-1)=f^2(f-1)$ and $b=cd$.

Remark. The discriminant of this family vanishes exactly where $f(f-1)(f^2-f+1)(f^3-6f^2+3f+1)=0$, and for the other $f\in k$ the k-point $(0,0)$ has order exactly 9 (checked in Magma).

(9) With d and f as in (8), suppose that $m\geqslant 9$, so that $5P,\dots,8P$ are not O. The following code takes $b=cd$, works over the field $\mathbb{Q}(c,d)$ of rational functions, and factors the numerators of $x(4P)-x(8P)$ and $x(5P)-x(6P)$ after the substitution $c=f(d-1)$.

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K<c, d> := FunctionField(Rationals(), 2);
E := EllipticCurve([1 - c, -c*d, -c*d, 0, 0]);    // b = c*d
P := E![0, 0];
4*P;                      // (d^2 - d : c*d^2 - d^3 + d^2 : 1)
R<f, d> := PolynomialRing(Rationals(), 2);
// the irreducible factors of the numerator of x(iP) - x(jP), with c = f(d - 1)
Factors := func<i, j | [g[1] : g in Factorization(
  Evaluate(Numerator((i*P)[1] - (j*P)[1]), [f*(d - 1), d]))]>;
Factors(4, 8);
// [ d, d - 1, f - 1, f^2 - f*d^2 - f*d + 3*d^2 - 3*d + 1 ]
Factors(5, 6);
// [ d - 1, f, f^3*d - 3*f^2*d + 4*f*d - f - d^2 ]
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Solution. For $i\neq j$ with $0<|i-j|<m$, the k-points iP and jP have the same x-coordinate if and only if $iP=-jP$, since $iP=jP$ would give $(i-j)P=O$. The numerators vanish exactly where the x-coordinates agree. Indeed the denominators of $x(5P)$, $x(6P)$ and $x(8P)$ are $(d-1)^2$, $(c-d+1)^2$ and $\big(cd-(2d-1)(d-1)\big)^2$ (checked in Magma), which vanish only where $5P$, $6P$ or $8P$ is O. For $8P$, this holds as $8P=O$ means that $4P=-4P$, that is, $2y(4P)+(1-c)x(4P)-cd=d\big(cd-(2d-1)(d-1)\big)=0$. And $d\neq 0$ (as $b\neq 0$), $d\neq 1$ (as $m\neq 5$), $f\neq 0$ (as $c\neq 0$) and $f\neq 1$ (as $f=1$ means $c=d-1$, that is, $b=c+c^2$ and $m=6$).

(a) Show that $m=12$ if and only if

$$f^2-(d^2+d)f+3d^2-3d+1=0.$$

Suppose that $\operatorname{char} k\neq 2$. Find a one-parameter family of elliptic curves with a k-point of order 12.

Solution. **Take $d=(v^2-2v+2)/v$ and $f=(3v^2-6v+4)/v^2$, with $v\in k^\times$.** Since $m\geqslant 9$, the x-coordinates of $4P$ and $8P$ agree if and only if $12P=O$, that is, $m=12$. By the solution above and the first line of output, this holds if and only if $f^2-(d^2+d)f+3d^2-3d+1=0$. Its discriminant as a polynomial in f is

$$(d^2+d)^2-4(3d^2-3d+1)=d^4+2d^3-11d^2+12d-4=(d-1)^2(d^2+4d-4).$$

So, for $d\neq 1$, there is a solution $f\in k$ if and only if $d^2+4d-4=w^2$ for some $w\in k$, that is, $(d+2-w)(d+2+w)=8$. Putting $d+2-w=2v$ and $d+2+w=4/v$, with $v\in k^\times$, gives

$$d=v+\frac{2}{v}-2=\frac{v^2-2v+2}{v},\qquad w=\frac{2}{v}-v,\qquad f=\frac{d^2+d+(d-1)w}{2}=\frac{3v^2-6v+4}{v^2},$$

the other root f corresponding to $2/v$ in place of v. Then $c=f(d-1)$ and $b=cd$.

Remark. The discriminant of this family vanishes exactly where

$$(v-1)(v-2)(3v^2-6v+2)(3v^2-6v+4)(v^2-2v+2)=0,$$

and for the other $v \in k^\times$ the k-point $(0,0)$ has order exactly 12 (checked in Magma).

(b) Show that $m=11$ if and only if

$$d^2-(f^3-3f^2+4f)d+f=0.$$

Suppose that $\operatorname{char} k \notin \{2,11\}$. Show that this curve is birational to the curve

$$w^2=f(f^3-4f^2+8f-4),$$

and that the latter has genus one.

Solution. **Put $w := \big(2d-(f^3-3f^2+4f)\big)/(f-1)$.** Since $m \geqslant 9$, the x-coordinates of $5P$ and $6P$ agree if and only if $11P=O$, that is, $m=11$, as 11 is prime. By the solution above and the second line of output, this holds if and only if $d^2-(f^3-3f^2+4f)d+f=0$. Completing the square,

$$\big(2d-(f^3-3f^2+4f)\big)^2=(f^3-3f^2+4f)^2-4f=f(f-1)^2(f^3-4f^2+8f-4).$$

As $f \neq 1$, dividing by $(f-1)^2$ gives $w^2=f(f^3-4f^2+8f-4)$, and $d=\big((f-1)w+f^3-3f^2+4f\big)/2$ recovers d, so the two curves are birational. The quartic $f(f^3-4f^2+8f-4)$ has distinct roots, since 0 is not a root of the cubic, whose discriminant $-176=-2^4\cdot 11$ is not 0 in k. So the smooth projective model C of $w^2=f(f^3-4f^2+8f-4)$ has a map $f\colon C \to \mathbb{P}^1$ of degree 2, ramified exactly over the four roots, and not over ∞, as the degree 4 is even. As $\operatorname{char} k \neq 2$, Hurwitz's formula [Sil09, Theorem II.5.9] is an equality, $2g-2=2(-2)+4=0$, and C has genus one.

Remark. By (1) and (3), the pairs (E,P) of an elliptic curve over k and a k-point P of order $m \geqslant 4$, up to change of variables, correspond bijectively to the $(b,c) \in k^2$ with $\Delta \neq 0$ that satisfy the condition for m. For $m=7,9,11,12$ these are the k-points of an affine curve $Y_1(m)$, in the coordinates d, f or v above, and the Kubert–Tate curve over $Y_1(m)$, with its section $(0,0)$, is a universal elliptic curve with a section of order m: every pair (E,P) over k is its fiber over exactly one k-point of $Y_1(m)$. The smooth projective model of $Y_1(m)$ is the modular curve $X_1(m)$, and the points added are the cusps [Sil09, Theorem C.13.1(b) and Remark VIII.7.8]. Example C.13.4 there is the family of (7), with $t=d$. For the orders $4 \leqslant m \leqslant 10$ and $m=12$ of Theorem 1.1, the elliptic curves with a k-point of order m lie in a one-parameter family [Sil09, Remark VIII.7.8], and $X_1(m)$ is rational. The order 11 is the first one left out, and $X_1(11)$ has genus one. Over $\mathbb{Q}$, Magma identifies C with the elliptic curve $y^2-y=x^3-x^2$ (Cremona's 11a3), of rank 0 with five rational points. The curve $X_1(11)$ has five rational cusps [Sil09, Example C.13.3]. So every rational point of $X_1(11)$ is a cusp, and no elliptic curve over $\mathbb{Q}$ has a rational point of order 11, as Theorem 1.1 says. In this language, Theorem 1.1 says that every rational point of $X_1(m)$ is a cusp for $m=11$ and $m \geqslant 13$ [Sil09, Remark C.13.5].

2. No good reduction everywhere

This section is [Sil09, Exercise 8.15], and follows its hint. A Weierstrass equation over $\mathbb{Z}$ with $\Delta=\pm 1$ would have good reduction at every prime, and congruences modulo 8 rule it out.

In the following definitions and results, E is an elliptic curve over $\mathbb{Q}$ and p is a prime number, with valuation v_p. Write $\mathbb{Z}_{(p)}$ for the ring of rational numbers whose denominators are prime to p.

Definition 2.1 (Minimal equations [Sil09, §VII.1])**.** A Weierstrass equation for E is minimal at p if its coefficients lie in $\mathbb{Z}_{(p)}$ and $v_p(\Delta)$ is as small as possible among such equations for E.

Proposition 2.2 (Good reduction [Sil09, Proposition VII.5.1(a)]). *The curve E has good reduction at p if and only if a Weierstrass equation for E minimal at p has $v_p(\Delta) = 0$.*

Corollary 2.3 (Global minimal equations [Sil09, Corollary VIII.8.3]). *The curve E has a Weierstrass equation with coefficients in $\mathbb{Z}$ that is minimal at every prime p.*

Proposition 2.4 (Weierstrass formulas [Sil09, §III.1]). *Put*

$$b_2 = a_1^2 + 4a_2, \qquad b_4 = 2a_4 + a_1a_3, \qquad b_6 = a_3^2 + 4a_6,$$
$$b_8 = a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2, \qquad c_4 = b_2^2 - 24b_4, \qquad c_6 = -b_2^3 + 36b_2b_4 - 216b_6.$$

Then $\Delta = -b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6$, *and* $1728\Delta = c_4^3 - c_6^2$.

Problem 2.5 (No good reduction everywhere).
Let E be an elliptic curve over $\mathbb{Q}$. Suppose, for a contradiction, that E has good reduction at every prime p.

(1) Show that E has a Weierstrass equation with coefficients in $\mathbb{Z}$ and $\Delta = \pm 1$. Fix such an equation, and put $\varepsilon := \Delta$.

Solution. **Take the equation of Corollary 2.3.** By Corollary 2.3, E has a Weierstrass equation with coefficients in $\mathbb{Z}$ that is minimal at every prime p. By Proposition 2.2, $v_p(\Delta) = 0$ for every p. As $\Delta \in \mathbb{Z}$, it is ± 1.

(2) Show that a_1 is odd.

Solution. **If a_1 were even, then $\Delta \equiv 5 \pmod 8$.** Suppose that a_1 is even. Then $b_2 = a_1^2 + 4a_2 \equiv 0 \pmod 4$ and $b_4 = 2a_4 + a_1a_3$ is even. So $b_2^2b_8 \equiv 0 \pmod{16}$, $8b_4^3 \equiv 0 \pmod 8$ and $9b_2b_4b_6 \equiv 0 \pmod 8$, and Proposition 2.4 gives $\Delta \equiv -27b_6^2 \pmod 8$. As Δ is odd, so is b_6, and then $b_6^2 \equiv 1 \pmod 8$ and $\Delta \equiv -27 \equiv 5 \pmod 8$. But $\pm 1 \equiv 1$ or $7 \pmod 8$, a contradiction.

(3) Show that $c_4 \equiv 1 \pmod 8$.

Solution. **The integer b_2 is odd, so $b_2^2 \equiv 1 \pmod 8$.** By (2), $b_2 = a_1^2 + 4a_2$ is odd, so $b_2^2 \equiv 1 \pmod 8$. And $24b_4 \equiv 0 \pmod 8$. So $c_4 = b_2^2 - 24b_4 \equiv 1 \pmod 8$.

(4) Put $u := c_4 - 12\varepsilon$. Show that

$$c_6^2 = u\,(u^2 + 36\varepsilon u + 432),$$

and deduce that u is a positive square or three times a positive square.

Solution. **The two factors are positive, and their greatest common divisor divides 27.** As $\varepsilon^2 = 1$, we have $c_4^3 = (u + 12\varepsilon)^3 = u^3 + 36\varepsilon u^2 + 432u + 1728\varepsilon$. By Proposition 2.4,

$$c_6^2 = c_4^3 - 1728\varepsilon = u\,(u^2 + 36\varepsilon u + 432).$$

Here $u^2 + 36\varepsilon u + 432 = (u + 18\varepsilon)^2 + 108 > 0$, and u is odd, by (3), so $u \neq 0$. As $c_6^2 \geqslant 0$, $u > 0$. A common divisor of u and $u^2 + 36\varepsilon u + 432$ divides $432 = 2^4 \cdot 3^3$, and is odd, so it divides 27. So every prime $\ell \neq 3$ divides at most one of the two factors, and $v_\ell(u) = v_\ell(c_6^2)$ is even. Hence $u = 3^e s^2$ for some $e \geqslant 0$ and some positive integer s prime to 3, and u is a square if e is even and three times a square if e is odd.

(5) Show that $u \equiv 5 \pmod 8$, and conclude that E has bad reduction at some prime p.

Solution. **Odd squares are 1 modulo 8, and three times an odd square is 3.** By (3), and as $12\varepsilon \equiv 4 \pmod 8$ for $\varepsilon = \pm 1$,

$$u = c_4 - 12\varepsilon \equiv 1 - 4 \equiv 5 \pmod 8.$$

But u is odd, so by (4) it is an odd square, which is 1 modulo 8, or three times an odd square, which is 3 modulo 8. This contradiction shows that E has bad reduction at some prime p.

Remark. By Shafarevich's theorem, for every finite set S of primes there are only finitely many elliptic curves over $\mathbb{Q}$, up to isomorphism, with good reduction at every prime outside S [Sil09, Theorem IX.6.1]. This problem is the case $S = \varnothing$, with none. For $S = \{2\}$ there are 24, and for $S = \{2, 3\}$ there are 784 [Sil09, Example IX.6.1.1].

References

[Kub76] D. S. Kubert, *Universal bounds on the torsion of elliptic curves*, Proc. London Math. Soc. (3) **33** (1976), no. 2, 193–237. ↑

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑